

What is the Small Area Demographic Estimates (SADE) model?

The SADE model from Washington State's Office of Financial Management (OFM) creates population numbers for small areas across the state. "Small areas" means places smaller than the whole state, which might not match up exactly with official boundaries used by the U.S. Census.

SADE gives population estimates broken down by age, sex, race, and ethnicity. It calculates numbers for very small places—down to the block level—but only shares results for larger areas.

What race category system does SADE use?

SADE follows federal Office of Management and Budget (OMB) rules from 1997 for grouping people by race and ethnicity. These rules include at least five race groups: White, Black or African American, American Indian and Alaska Native (AIAN), Asian, and Native Hawaiian or Other Pacific Islander (NHOP). There's also a group for people who identify with more than one race.

SADE uses Modified Age & Race Census (MARC) data to sort people who chose "some other race" into one or more of the five OMB groups.

How are the SADE estimates made?

SADE combines data from the Census and OFM for different places and groups. It uses a math process called Iterative Proportional Fitting (IPF) to make sure the numbers for age, sex, and race add up correctly.

First, it gathers age, sex, and race data from several Census sources. Then, it combines these into a baseline estimate and uses OFM data to track changes over time. It adjusts for things like unusual age patterns and population corrections. Finally, it uses IPF to make sure everything fits together, matching totals from OFM's other programs including: Small Area Estimates Program (SAEP), OFM's county level population estimates by age and sex, and OFM's county level estimates by race and ethnicity.

How do SADE estimates relate to official city and county numbers?

SADE's yearly estimates are matched to OFM's official April 1 population numbers at the county level. SADE numbers are not the official counts used for things like funding or programs. For official city and county numbers, visit the April 1 Population Estimates Program website:

<http://www.ofm.wa.gov/pop/april1/default.asp>.

What are the limitations of using SADE data?

SADE is based on block-level SAEP data, so any limits in that data apply to SADE too.

SADE uses proportions from the 2020 Census, which don't show changes like aging or moving in small areas. SADE can't be used to study these trends in detail.

Because the data is for small areas, numbers are fractions (like 0.5 people). These should be added up and rounded for bigger areas.

How small can the geography be for SADE estimates?

It's best to use SADE data for areas about the size of a census tract (around 5,000 people), using five-year age groups, six race categories (including multiracial), and two Hispanic categories. There are two main reasons:

- (1) Some details come from larger areas, so applying them to smaller places is based on modeling, not direct measurement.
- (2) With many groups (age, sex, race, etc.), the numbers in each group can get very small and unstable. For example, if 5,000 people are split evenly into all available groups, groups might have less than 10 people, which isn't reliable.

Custom areas don't have to be census tracts, but should have similar or larger populations. You can also combine groups (age, sex, race, etc.) to make the data more stable.

Why are there decimals, like 0.5 people?

OFM keeps numbers as decimals so totals match up correctly and to remind users these are estimates, not exact counts. See above entry regarding "limitations".

Are small numbers in the estimate "identifiable"?

Some groups in the data have very small numbers (for example Asian Hispanic 40-45 males in Adams County), but these are based on models and not actual people, so no individual can be identified.

Should I seek out other sources of regional and small area data?

SADE is meant for statewide use. It's not a replacement for local estimates used in regional planning. County and regional planning agencies can adjust their data for their area. If you're working within a local agency's area, contact them first.

How can I get SADE data?

SADE data is available on OFM's website at <http://www.ofm.wa.gov/pop/asr/default.asp>.

What kinds of SADE data are available?

SADE provides yearly data from 2000 onward for the state, counties, congressional districts, legislative districts, school districts, and census tracts.

There are three tables for each area: total population, Hispanic population, and non-Hispanic population. Data is grouped by race, sex, and five-year age groups

How often is SADE updated?

SADE is updated every year in the first quarter. Data before 2020 stay the same, but newer estimates are updated annually.

What if I need data for a different area or age group?

Contact OFM for custom data. If possible, they may provide it (sometimes for a fee) and may also make it public if it seems broadly useful.

Is there a suggested citation?

No specific style is required. Use your organization's usual style and mention the SADE data layer and release. For example in APA it looks like this:

Washington State Office of Financial Management, Forecasting Division (2024). Small Area Demographic Estimates: Census Tracts [Data file]. Retrieved from https://ofm.wa.gov/sites/default/files/public/dataresearch/pop/asr/sade/ofm_pop_sade_tract_2020_to_2023.xlsx.

Is there a liability disclaimer?

Yes. This disclaimer applies to many OFM data products, as well.

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